

Aadya Singh

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Profile

Master of Computer Science student at the University of Sydney specialising in Data Science, AI, and Software Engineering. Builds production-shaped AI and data systems across Python, FastAPI, PostgreSQL, Docker, Next.js, RAG, MLOps, and monitoring workflows. Public work spans GitHub projects, Kaggle datasets, and technical writing.

Education

Master of Computer Science (Advanced Entry)

University of Sydney

Specialisations: Data Science & AI, Software Engineering | Current WAM: 80

2025 - 2027

Sydney, Australia

PG Certificate in AI/ML & Data Science

Indian Institute of Technology Roorkee

Final Score: 85/100 | Machine Learning: 86 | Deep Learning: 84

2024 - 2025

Roorkee, India

Bachelor of Engineering in Biotechnology

M.S. Ramaiah Institute of Technology

First Class with Distinction | CGPA: 8.0/10

2019 - 2023

Bengaluru, India

Selected Projects

AgentOps Support Automator

Self-directed

2026

Python, FastAPI, LangGraph,

PostgreSQL, Docker, Next.js

- Built a multi-agent technical support workflow with dedicated ingestion, planning, resolution, and human-review stages.
- Implemented typed workflow state, approval gating, failure handling, retry paths, and API endpoints for inspecting task progress.
- Containerised the backend/frontend stack and added focused tests for agent routing and workflow behavior.

Code: github.com/aadyasingh55/agentops-support-automator

MLOps Drift Monitor

Self-directed

2026

Python, FastAPI, scikit-learn,

Evidently, PostgreSQL, Docker

- Built a predictive-maintenance API around the AI4I 2020 dataset with request logging, prediction history, and model metadata.
- Trained and served a scikit-learn classifier for machine-failure prediction while avoiding target leakage from failure-type labels.
- Added a drift-monitoring workflow to compare live feature distributions against reference data and surface monitoring reports.

Code: github.com/aadyasingh55/mlops-drift-monitor

Medical Retrieval-Augmented Generation System

Self-directed

2026

Python, Streamlit, PyTorch, Hugging

Face, SentenceTransformers,

scikit-learn

- Built a local medical RAG workflow for privacy-sensitive document analysis, covering ingestion, chunking, indexing, evidence retrieval, grounded generation, and evaluation.
- Improved Precision@3 from 66.7% to 83.3% with all-mpnet-base-v2, a 24.9% uplift over a TF-IDF baseline while maintaining 100% Recall@3.
- Benchmarked TF-IDF, all-mpnet-base-v2, and Bio ClinicalBERT retrievers through a Streamlit evaluation interface.

Code: github.com/aadyasingh55/Medical-RAG-System

Selected Projects Continued

Australian Rental Insight Copilot

Self-directed

2026

Python, Streamlit, pandas, Plotly, scikit-learn

- Built a rental-market analytics app with automated exploratory analysis, pricing insights, visualisation, and anomaly detection on Australian housing datasets.
- Integrated national listings and NSW rental bond data with schema inference, multivariable analysis, and Isolation Forest-based anomaly detection.
- Developed an interactive dashboard for suburb-level trend analysis and affordability comparison.

Code: github.com/aadyasingh55/Australian-Rental-Insight-Copilot

Open-source Documentation Contributions

Public pull requests

2026

GitHub, Markdown, technical writing

- Opened external pull requests to LangGraph and Buildkite documentation repositories, focusing on precise developer-facing wording.
- Practised contributor workflows including issue assignment, branch hygiene, maintainer communication, and documentation review.

PRs: github.com/aadyasingh55

Experience

Neuberg Anand Academy of Laboratory Medicine

Laboratory Intern

Aug 2022 - Sep 2022

Bengaluru, India

- Observed reference-laboratory workflows across diagnostic departments and gained exposure to clinical testing operations.

Apex Biotechnology Training and Research Institute

Bioinformatics Intern

Nov 2021

Chennai, India

- Completed short-term training in bioinformatics, computational analysis, and biological data interpretation.

Technical Skills

Languages	Python, TypeScript, JavaScript, SQL, R
Backend	FastAPI, SQLAlchemy, PostgreSQL, JWT auth, REST APIs, Pydantic
Frontend	React, Next.js, Tailwind CSS, Streamlit, Plotly
AI/ML	NLP, Transformers, RAG, LLMs, scikit-learn, TensorFlow/Keras, PyTorch, Evidently
Tools	Git, Docker, pandas, LangGraph, LangChain, LaTeX, GraphPad Prism

Achievements

- GATE 2025 - 92 percentile nationally in Data Science & AI.
- Kaggle Datasets Expert - global rank 520.
- Authored 10+ technical articles on Analytics Vidhya with multiple top-10 blogathon rankings.
- Opened public pull requests to LangGraph and Buildkite documentation repositories.
- IELTS Band 8.0.